

B.Sc. 1st Semester (Honours) Examination-2022-23

MATHEMATICS

Course ID : 12114 Course Code : SH/MTH/103/GE-1

**Course Title : Calculus, Geometry &
Vector Analysis (New)**

Time : 2 Hours

Full Marks : 40

The figures in the right hand margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Notations and symbols have their usual meaning.

Unit-I

1. Answer any five questions :

2×5=10

(a) Find the co-ordinate of the points where the origin is to be shifted so that the equation $3x^2+8xy+3y^2-2x+2y-2=0$ can be reduced to one which is free from linear terms.

(b) Evaluate $\lim_{x \rightarrow 0} (1 + \sin x)^{\cot x}$.

(c) Find the n^{th} order derivative of $x^3 \log x$.

(Turn Over)

- (d) Find $\int_0^{\frac{\pi}{2}} \sin^8 x \, dx$.
- (e) Find the volume of the solid generated by the revolution of the area enclosed by the asteroik $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ about the x axis.
- (f) show that $[\vec{\alpha} \times \vec{\beta} \, \vec{\beta} \times \vec{\gamma} \, \vec{\gamma} \times \vec{\alpha}] = [\vec{\alpha} \, \vec{\beta} \, \vec{\gamma}]^2$.
- (g) For the curve $\vec{r} = 2a \cos t \, \hat{i} + 2a \sin t \, \hat{j} + bt^2 \hat{k}$, show that $[\dot{\vec{r}} \, \ddot{\vec{r}} \, \dddot{\vec{r}}] = 8a^2bt$.
- (h) Find the radius and polar coordinates of the centre of the circle $r = 4\cos\theta + 3\sin\theta$.

Unit-II

2. Answer any *four* questions :

4×5

- (a) Give the definition of rectilinear asymptote and determine the asymptotes of the following algebraic curve $x^3 + x^2y - xy^2 - y^3 + 2xy + 2y^2 - 3x + y = 0$.

1+4

- (b) If $\log y = \tan^{-1}x$ then show that

$$(1 + x^2)y_{n+2} + (2nx + 2x - 1)y_{n+1} + n(n+1)y_n = 0.$$

- (c) Reduce the equation $4x^2 - 4xy + y^2 + 2x - 26y + 9 = 0$ to its suitable form to determine the conic.
- (d) A sphere of constant radius r passes through the origin O and cuts the axes in A, B, C . Prove that the locus of the foot of the perpendicular from O to the plane ABC is given by $(x^2 + y^2 + z^2)^2(x^2 + y^2 + z^2) = 4r^2$.
- (e) (i) Prove a necessary and sufficient conditions that a proper vector $\vec{u}$ has a constant length is that

$$\vec{u} \frac{d\vec{u}}{dt} = 0.$$

- (ii) If $\vec{a} = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ then find the value of

$$\vec{a} \times \frac{d^2\vec{a}}{dt^2} \text{ at } t = 1. \quad 3+2$$

- (f) (i) Evaluate $\int_2^3 \left(\vec{r} \times \frac{d^2\vec{r}}{dt^2} \right) dt$ where $\vec{r} = t^3\hat{i} + 2t^2\hat{j} + 3t\hat{k}$.

- (ii) If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ be coplanar vectors, then show that

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}. \quad 3+2$$

Unit-III

3. Answer any one question :

1×10

(a) (i) If $y = (x^2 - 1)^n$, then show that

$$(x^2 - 1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0.$$

(ii) Find the volume of the solid generated by revolving the cardioids $r = a(1 - \cos\theta)$ about the initial line.

(iii) Find the point of inflexion of the curve $x = (\log y)^3$, if any.

4+3+3

(b) (i) State and prove the reflection property of parabola.

(ii) If $I_n = \int \sin^n \theta d\theta$ then show that $nI_n = \sin \theta \cos \theta - (n-1)I_{n-2}$.

(ii) Prove that $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{b} \times \vec{a}$ intersect and find their intersection.

4+3+3

**Course Title : Calculus, Geometry &
Differential Equation (Old)**

Unit-I

1. Answer any five questions :

2×5=10

(a) Find the co-ordinate of the points where the origin is to be shifted so that the equation $2x^2 - 3xy + 4y^2 + 10x - 19y + 23 = 0$ can be reduced to one which is free from linear terms.

(b) If $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x dx$ and $n > 1$, show that

$$I_n + n(n-1)I_{n-2} = n(\pi/2)^{n-1}.$$

(c) Prove that the curve $y = e^{2x}$, is always convex to x axis at every point.

(d) If $\lim_{x \rightarrow c} \{f(x)/h(x)\}$ exists and is finite and if $\lim_{x \rightarrow c} h(x) = 0$ then prove that necessarily $\lim_{x \rightarrow c} f(x) = 0$.

(e) Show that $y = 3(x+1)$ is an oblique asymptote of the curve $y = 3x + \frac{3x}{x-1}$.

- (f) Solve $\frac{dy}{dx} = (x^2 + 1)(y^2 + 1)$.
- (g) Find an integrating factor of $(x^2 + y^2 + 2x)dx + 2y dy = 0$.
- (h) Find the polar equation of parabola whose latus rectum is 6.

Unit-II

2. Answer **any four** questions : 5×4=20

- (a) Reduce the equation $x^2 + 4xy + y^2 - 2x + 2y + 6 = 0$ to its canonical forms and determine the nature of the conic.
- (b) (i) If $y = \cos (10 \cos^{-1} x)$ then show that
 $(1 - x^2)y_{12} - 21xy_{11} = 0$.
- (ii) If $y = \sin^3 x$ then find $\frac{d^n y}{dx^n}$.
- (c) Prove that the asymptotes of the curve $x^2y - y^3 - 2ay^2 + 5x - 7 = 0$ forms a triangle of area a^2 .
- (d) Find the equation of the sphere touching the three co-ordinate planes.
- (e) Show that the equation of the tangent to the conic
 $\frac{l}{r} = 1 + e \cos \theta$ parallel to the tangent at $\theta = \alpha$ is given
 by $l(e^2 + 2e \cos \alpha + 1) = r(e^2 - 1)[\cos(\theta - \alpha) + e \cos \theta]$.

(f) Solve : $\frac{dy}{dx} + y \cos x = y^n \sin 2x$.

Unit-III

3. Answer **any one** question :

10×1=10

(a) (i) Define generating line of the surface. Find the equation of the generators of the hyperboloid $x^2 - y^2 = 4z$, which passes through the point (7,5,6).

(ii) Find a and b in order that $\lim_{x \rightarrow 0} \frac{a \sin 2x - b \sin x}{x^3} = 1$.

(iii) If $f(x)=x^m$ then prove that n th ($n>m$) order derivative of $f(x)$ is zero. 5+3+2

(b) (i) Prove that the envelope of the family of circle passing through the origin and having centres situated on the hyperbola $x^2 - y^2 = c^2$ is $(x^2 + y^2)^2 = 4c^2 (x^2 - y^2)$.

(ii) Find the general solution and singular solution of the differential equation $y = px + \sqrt{1 + p^2}$.

(iii) Solve the differential equation

$$\frac{dy}{dx} + yf'(x) = f(x)f'(x). \quad 4+4+2$$

Internal Assessment, 2022

Sub: Math, GE Sem: I

Course: Calculus, Geometry and Vector.

Answer any five questions:-

$$\underline{5 \times 2 = 10}$$

- (1) For what value of a the straight line $3x + 4y = 1$ touches the parabola $y^2 = 4ax$?
- (2) Find $\int \frac{dx}{(1+x^2)\sqrt{1+x^2}}$.
- (3) Prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$.
- (4) If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 3$, then find the angle between $\vec{a}$ and $\vec{b}$.
- (5) Find $\int \cos^5 x dx$.
- (6) Find the area bounded by the parabola $y^2 = 16x$ and the st. line $x = 4$.
- (7) Find the centre of the circle $r = \cos \theta$.
- (8) The length of the latus rectum of the parabola $x^2 + 8x + 4y + 16 = 0$.